Residual Multichannel Budgets after Dispersion Transfer

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Abstract

We recompute the normalized defect budget after isolating the restricted continuous dispersion contribution and auditing the remaining sector channels.

Let q_* be the total ceiling and suppose certified channels contribute $q_C < q_*$. For open channels c choose positive shares α_c summing to one.

Theorem 1. *The exact residual allocation is*

$$q_c = \alpha_c(q_* - q_C), \qquad F_c \geq \ell_c - \log q_c.$$

The channel union is at most q_* whenever every certified premise and every open-channel numerator bound holds uniformly over the same retained state.

Proof. Subadditivity gives the union bound. The residual identity follows from $\sum_c \alpha_c = 1$, and substitution into $e^{\ell_c - F_c} \leq q_c$ gives the free-energy conditions. □

The restricted continuous calculation leaves a positive residual budget for center, cut/orbit, and integer-topology channels.

Remark 1. *This is not yet a physical union bound: the continuous premises and all three remaining channel ratios are still nonuniform or open. The single-star repair is excluded by Stage 44.*